

Alireza Shafaei

 Alireza-Shafaei |  shafae.erfan@yahoo.com |  shafaei.erfan.20019@gmail.com |  (+98) 991 263 0612 |
 No. 82, West 124 Alley, Tehran Pars, Tehran, Iran
Date of Birth: 11 Sep. 2001 | Gender: Male

EDUCATION

- Sep. 2024 – Current **M.S. in Electrical Engineering (Digital Systems)**, Sharif University of Technology
GPA: 18.02/20
- Sep. 2019 – Sep. 2024 **B.S. in Electrical Engineering**, Sharif University of Technology
GPA: 17.03/20
- Feb. 2021 – Sep. 2024 **Minor in Economics**, Sharif University of Technology

RESEARCH INTERESTS

Generative Modeling; Machine Learning; Deep Learning; Large Language Models & RAG; Reinforcement Learning; Mathematical Optimization Theory; Edge-Cloud Computing; Neuroscience; Embedded Systems; Economics.

SELECTIVE COURSES

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|------------------------------------|-------------------------------------|--|
| – Deep Generative Models (MSc) | – Memory, Cognition (MSc) | – Design Microprocessor Systems |
| – Statistical Learning (MSc) | – Adv. Computer Architecture (MSc) | – Econometrics |
| – Big Data Analytics (MSc) | – Intro. to Machine Learning | – Financial Economics |
| – Convex Optimization 2 (MSc) | – Convex Optimization 1 | – Digital Economics |
| – Deep Learning (MSc) | – Principles of Solid State Devices | – Macroeconomics |
| – Neural Networks (MSc) | – Communication Circuits | – Microeconomics |
| – Data communication network (MSc) | – Electronics 3 | – System-2 AI (MSc, Audit) |
| – Embedded Systems (MSc) | – Pulse Tech. and Digital Circuits | – Deep Reinforcement Learning (MSc, Audit) |
| – Neuroscience of Learning, | | |

ACADEMIC PROJECTS

- **LLM-Enabled In-Context Learning for Task Offloading in Edge Networks (2025-present):** Resource allocation as a service (DNN Partitioning) utilizing RAG and Prompting techniques, integrated with mathematical optimization (queueing stability and Lyapunov drift). *Supervisor: Dr. Shah Mansouri.*
- **Reinforcement Learning and Applications (2024):** Using Q-Learning to train agents in continuous environments; reducing resolution by extracting important states. *Supervisor: Dr. Bagheri Shouraki.*

SKILLS

Programming	Python, PyTorch, PySpark, Tensorflow, scikit-learn, Django, C, Verilog, MATLAB
Design & Sim	PSpice, HSPICE, Proteus, ADS, Omnet++
Other	MIPS/AVR/ARM, Linux, Embedded Systems, L ^A T _E X, Microsoft Office, Problem Solving

COURSE PROJECTS

- **Generative Models & Deep Learning (2025–2026):**
 - Implemented and trained an Energy-Based Model (EBM) on the MNIST dataset using Langevin dynamics.

- Developed a time-aware post-training quantization framework for diffusion language models to achieve significant model compression and inference acceleration.
- Implemented WaveNet (autoregressive generative model for raw audio), Probabilistic Graph Forecasting, CVAE for digit generation, and normalizing flows (RealNVP, PermutFlow).
- **Advanced Machine Learning & Data Analysis (2024–2025):**
 - *Massive Data Analysis:* Designed graph/collaborative recommenders (Pixie-style, Topic-Specific PageRank, HITS) and built a PySpark Structured Streaming pipeline for real-time sentiment aggregation.
 - *Statistical Learning:* Built end-to-end pipelines for age/gender estimation (Linear/Ridge/Lasso, PCA, ROC-AUC) with a custom drag-and-drop Python GUI for bounding box face detection.
 - Implemented GAN-BERT for AI vs. Human text recognition, complex CNNs for TIG Welding classification, and Hopfield/SOM networks.
- **Networking & Embedded Systems (2024–2025):**
 - *Data Networks:* Implemented a UDP-based DNS server and verified queries in Wireshark; built socket-level SMTP clients and a mailbox-capable SMTP server with auth.
 - *IoT Real-time Event Detection (Orange Pi):* Developed an edge device with LAMP dashboard, Python audio detectors, and MQTT pub/sub to a Mosquitto broker, hardened with systemd limits.
 - Implemented Data Link, MAC, and Network layers in Omnet++.
- **Hardware & Circuit Design (2019–2023):** Designed and simulated MIPS multi-cycle architecture (Verilog), high-frequency MOSFET Amplifiers and Opamp 741 (HSPICE), digital stopwatches, and multivibrators (Proteus/PSpice).
- **Early ML & Computational Modeling (2023–2024):** Hodgkin–Huxley & LIF neuron simulations, VAE/CVAE implementations, clustering with EM Algorithm for GMMs, and classical classification (SVM, Decision Trees, AlexNet Fine-tuning).

TEACHING ASSISTANTSHIP

- Neural Networks (MSc) — Dr. Bagheri Shouraki (2025)
- Machine Learning and Vision Lab — Dr. Mohammadzadeh (2025)
- Electronics 1 & 2 — Dr. Fakharzadeh (2022, 2023)
- Principles of Solid State Devices — Dr. Sarvari (2023)
- Design Microprocessor Systems — Dr. Hajsadeghi (2023)

HONORS & AWARDS

- Ranked **21st** among 20,000 in Iranian University Entrance Exam for M.Sc. in Electrical Engineering, 2024
- Ranked **24th** among 150,000 in Iranian University Entrance Exam (Math & Physics), 2019

LANGUAGES

Persian Native
English Advanced